# THE 2-DOMINATION NUMBER AND THE UPPER MEDIAN DEGREE: A PROOF OF GRAFFITI.PC CONJECTURE 387

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ABSTRACT. Let $G$ be a nonempty finite simple graph of order $n$, and let $m(G)$ be the upper median of its degree sequence. We prove that

$$\gamma_2(G) \leq n - m(G) + 1,$$

where $\gamma_2(G)$ is the minimum cardinality of a set $D \subseteq V(G)$ such that every vertex outside $D$ has at least two neighbors in $D$. This confirms Graffiti.pc Conjecture 387. In fact, the argument shows that the connectedness hypothesis in the original formulation is unnecessary. The proof uses the complement graph and a minimally linearly dependent family of polynomials encoding selected non-neighborhoods.

## 1. INTRODUCTION

All graphs in this note are finite and simple. A set $D \subseteq V(G)$ is *2-dominating* if every vertex in $V(G) \setminus D$ has at least two neighbors in $D$. The minimum cardinality of such a set is the *2-domination number* $\gamma_2(G)$. For general background on domination in graphs, see [2].

Let the degree sequence of an $n$-vertex graph $G$ be written in nondecreasing order as

$$d_1 \leq d_2 \leq \cdots \leq d_n.$$

Its *upper median degree* is

$$m(G) = d_{\lfloor n/2 \rfloor + 1}.$$

A conjecture generated by Graffiti.pc and recorded by DeLaViña, Larson, Pepper, and Waller [1, Conjecture 1] predicts, for connected graphs, the upper bound

$$\gamma_2(G) \leq n - m(G) + 1. \tag{1}$$

The displayed statement of that conjecture in [1] contains the opposite inequality sign; however, the sentence immediately preceding it explicitly states the upper bound (1), and the subsequent discussion and partial results use the upper-bound direction. Thus the displayed sign is a typographical error.

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We prove the conjectured inequality for every nonempty finite simple graph, without assuming connectedness.

**Theorem 1.** *Let $G$ be a nonempty finite simple graph of order $n$. Then*

$$\gamma_2(G) \leq n - m(G) + 1.$$

The only algebraic ingredient is the following elementary lemma about a minimal linear dependence among polynomials.

## 2. A polynomial lemma

**Lemma 1.** *Let $B$ be a finite set, let $t \geq 0$, and choose pairwise distinct real numbers $a_z$ for $z \in B$. Suppose $A_1, \ldots, A_r$ are $t$-element subsets of $B$ and*

$$p_i(X) = \prod_{z \in A_i} (X - a_z) \qquad (1 \leq i \leq r).$$

*Assume that the indexed family $(p_i)_{i=1}^r$ is minimally linearly dependent over $\mathbb{R}$. Put*

$$P = \bigcap_{i=1}^r A_i, \qquad s = |P|.$$

*Then*

1. *$2 \leq r \leq t + 2$;*
2. *every element of $B \setminus P$ belongs to at most $r - 2$ of the sets $A_1, \ldots, A_r$;*
3. *$s \leq t + 2 - r$.*

*Proof.* Each $p_i$ is nonzero, so $r \geq 2$. Minimal dependence implies that any $r - 1$ of the polynomials are linearly independent. Since all $p_i$ lie in the $(t + 1)$-dimensional vector space of real polynomials of degree at most $t$, we obtain $r - 1 \leq t + 1$, and hence $r \leq t + 2$.

Choose a nontrivial dependence

$$\sum_{i=1}^r c_i p_i = 0. \tag{2}$$

Minimality implies $c_i \neq 0$ for every $i$. If some $z \in B$ belonged to exactly $r - 1$ of the sets $A_i$, then evaluating (2) at $X = a_z$ would leave exactly one nonzero summand. Indeed, if $z \notin A_j$, then

$$p_j(a_z) = \prod_{u \in A_j} (a_z - a_u) \neq 0$$

because the numbers $a_u$ are pairwise distinct. This contradiction shows that an element occurring in at least $r - 1$ of the sets must occur in all $r$ of them. Consequently, every element outside $P$ occurs in at most $r - 2$ sets.

Finally, define

$$g(X) = \prod_{z \in P} (X - a_z)$$

and write $p_i = g\widetilde{p}_i$. Multiplication by the nonzero polynomial $g$ is injective, so the indexed family $(\widetilde{p}_i)_{i=1}^r$ is again minimally linearly dependent. Each $\widetilde{p}_i$ has degree $t-s$. Therefore any $r-1$ of them are linearly independent in a vector space of dimension $t-s+1$, giving

$$r - 1 \le t - s + 1.$$

Equivalently, $s \le t + 2 - r$. □

## 3. Proof of the main theorem

*Proof of Theorem 1.* Write $m = m(G)$. If $m = 0$, then $V(G)$ is a 2-dominating set and

$$\gamma_2(G) \le n < n + 1 = n - m + 1.$$

Hence assume $m \ge 1$. Let $F = \overline{G}$ and put

$$t = n - 1 - m, \qquad q = t + 2 = n - m + 1.$$

In particular, $0 \le t \le n-2$ and $2 \le q \le n$.

Define

$$H = \{x \in V(G) : d_F(x) \le t\}, \qquad B = V(G) \setminus H,$$

and write $h = |H|$ and $k = |B|$. Since

$$d_F(x) = n - 1 - d_G(x),$$

a vertex lies in $H$ exactly when its degree in $G$ is at least $m$. By the definition of the upper median, at least $\lceil n/2 \rceil$ vertices have degree at least $m$. Thus

$$h \ge \left\lceil \frac{n}{2} \right\rceil, \qquad k \le \left\lfloor \frac{n}{2} \right\rfloor, \qquad h \ge k. \tag{3}$$

First suppose that $k \le t+1$. Since $k < q \le n$, choose a $q$-element set $D$ with $B \subseteq D$. Every vertex $v \notin D$ belongs to $H$, so

$$d_F(v, D) \le d_F(v) \le t.$$

As $v \notin D$, its neighbors in $F$ among the vertices of $D$ are exactly its nonneighbors in $G$ among those vertices. Hence

$$d_G(v, D) = |D| - d_F(v, D) \ge q - t = 2.$$

Thus $D$ is 2-dominating.

It remains to consider the case

$$k \ge t + 2. \tag{4}$$

For each $x \in H$, we have

$$|N_F(x) \cap B| \le d_F(x) \le t.$$

Using (4), extend $N_F(x) \cap B$ to a $t$-element subset $A_x \subseteq B$. Choose pairwise distinct real numbers $a_z$ for $z \in B$, and define

$$p_x(X) = \prod_{z \in A_x} (X - a_z) \qquad (x \in H),$$

where the empty product is 1 when $t = 0$.

The vector space of real polynomials of degree at most $t$ has dimension $t+1$. By (3) and (4),

$$h \geq k \geq t+2,$$

so the indexed family $(p_x)_{x \in H}$ is linearly dependent. Choose an inclusion-minimal dependent subfamily, indexed by a set $C \subseteq H$, and put $r = |C|$. Apply Lemma 1 to the sets $(A_x)_{x \in C}$. With

$$P = \bigcap_{x \in C} A_x, \qquad s = |P|,$$

the lemma gives

$$s \leq t + 2 - r, \tag{5}$$

and every element of $B \setminus P$ belongs to at most $r - 2$ of the sets $A_x$ with $x \in C$.

Because $2 \leq r \leq t+2$, we have $0 \leq t + 2 - r \leq t$. Together with (5) and $|B| = k \geq t + 2$, this allows us to choose $Y \subseteq B$ such that

$$P \subseteq Y, \qquad |Y| = t + 2 - r.$$

Set

$$D = C \cup Y.$$

Since $C \subseteq H$ and $Y \subseteq B$, the union is disjoint and

$$|D| = r + (t + 2 - r) = t + 2 = q.$$

Let $v \notin D$. If $v \in H$, then simply

$$d_F(v, D) \leq d_F(v) \leq t.$$

Now suppose $v \in B$. Since $P \subseteq Y$ and $v \notin Y$, we have $v \notin P$. Lemma 1 therefore shows that $v$ belongs to at most $r - 2$ of the sets $A_x$, $x \in C$. As $N_F(x) \cap B \subseteq A_x$ for every $x \in C$, and $F$ is undirected,

$$d_F(v, C) = |\{x \in C : v \in N_F(x) \cap B\}| \leq r - 2.$$

Consequently,

$$d_F(v, D) = d_F(v, C) + d_F(v, Y) \leq (r - 2) + |Y| = t.$$

We have shown that every $v \notin D$ satisfies $d_F(v, D) \leq t$. Therefore

$$d_G(v, D) = |D| - d_F(v, D) \geq (t + 2) - t = 2.$$

Thus $D$ is a 2-dominating set of cardinality

$$|D| = q = n - m + 1,$$

and the theorem follows. □

*Remark* 1. The bound is sharp for every $n \geq 2$: for the complete graph $K_n$, one has $m(K_n) = n - 1$ and $\gamma_2(K_n) = 2$.

*Remark* 2. Connectedness is not used anywhere in the proof. Hence the conclusion strictly strengthens the original formulation of Graffiti.pc Conjecture 387.

## References


[1] E. DeLaViña, C. E. Larson, R. Pepper, and B. Waller, *Graffiti.pc on the 2-domination number of a graph*, Congressus Numerantium **203** (2010), 15–32.

[2] T. W. Haynes, S. T. Hedetniemi, and P. J. Slater, *Fundamentals of Domination in Graphs*, Monographs and Textbooks in Pure and Applied Mathematics, vol. 208, Marcel Dekker, New York, 1998.